

Primordial helium bounds on a leptonic electron-mass transition inside the nucleosynthesis window

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We treat the amplitude ε of a universal fractional shift in the electron mass that turns on inside the big-bang nucleosynthesis (BBN) temperature window as a free parameter, and ask what the measured primordial helium mass fraction Y_p implies for it. The shift is switched on below a critical temperature T_c by a linear ramp, $\varepsilon_{\text{eff}}(T) = \varepsilon \max(0, 1 - T/T_c)$, which deforms the weak rates in the nuclear network without changing the relativistic degrees of freedom. At the T_c where the response was measured in a production BBN code ($T_c \simeq 179$ keV), the helium response is linear, $dY_p/d\varepsilon = 0.00163$ per percent of ε , and reproduces a direct windowed run to a few percent of the window itself. Against the Aver *et al.* determination $Y_p = 0.2453 \pm 0.0034$, that elasticity implies

$$\varepsilon < 3.2\% \quad (2\sigma).$$

The EMPRESS determination cannot be used for the same purpose: standard BBN with $\varepsilon = 0$ already sits 2.9σ above it, so a measurement in tension with the null hypothesis does not constrain a deformation of that null. Deuterium is not used for a derivative bound, because its ε -response is a nonlinear bottleneck quantity. The result is a genuine upper limit on the amplitude of an electron-mass transition that switches on inside the nucleosynthesis window, with zero fitted parameters, and no dependence on any unconverged cosmological posterior.

I. INTRODUCTION

A fractional change in the electron mass at early times is a standard deformation of big-bang nucleosynthesis and of recombination. In the microwave background it is partly degenerate with the expansion history and has been studied as a possible contribution to the Hubble tension [1–3]. At nucleosynthesis the same deformation acts through the weak rates and the binding-sensitive nuclear network, and the response matrices for varying constants have long been tabulated [4].

Most existing bounds assume a constant shift from some high temperature through the BBN window, or a power-law variation in redshift. The case of interest here is different: an electron-mass shift that is *absent* above a critical temperature T_c and turns on below it. That topology matters because neutron–proton freeze-out and the deuterium bottleneck sit at different temperatures (~ 800 keV and ~ 70 keV, respectively). A transition that switches on between them deforms the two epochs unequally.

We do not propose a mechanism for such a transition, and we do not fix T_c or ε to any particular model value. The question is purely kinematic: if an electron-mass shift of amplitude ε turns on inside the BBN window, what does the measured helium mass fraction allow? The answer is a single number against one observational determination, and an explicit statement that the other common determination cannot be used for this purpose.

Section II fixes the ramp and the free parameters. Section III reports the measured helium response. Sec-

tion IV derives the Aver bound. Section V explains why EMPRESS yields no bound. Section VI states what is deliberately left out. Section VII summarises.

II. SETUP

Let $m_e^{(0)}$ be the present electron mass. We write a temperature-dependent fractional shift

$$m_e(T) = m_e^{(0)} [1 + \varepsilon_{\text{eff}}(T)], \quad (1)$$

with a linear turn-on below a critical temperature T_c ,

$$\varepsilon_{\text{eff}}(T) = \varepsilon \max(0, 1 - T/T_c). \quad (2)$$

Above T_c the shift is off; at $T \rightarrow 0$ it saturates at the constant amplitude ε . Both ε and T_c are free. The only structural commitment is the ramp shape of Eq. (2), which is the minimal continuous switch-on that vanishes above T_c and is linear in the order-parameter sense below it. Nothing below depends on why the switch-on has that form.

The relevant temperature window for a transition that BBN can actually witness is bounded below by the deuterium bottleneck (~ 70 keV) and above by the weak-rate window approaching neutron–proton freeze-out (of order a few hundred keV). We adopt the interval

$$T_c \in [70, 500] \text{ keV} \quad (3)$$

as the range over which T_c is regarded as free for the present purpose. Outside that interval the transition either misses the bottleneck entirely (too low) or reaches freeze-out in full (too high); both extremes change the

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interpretation of a helium-only bound and are not the case studied here.

The shift enters BBN through the electron mass in the weak rates. It does not add relativistic degrees of freedom: a network run with the ramp at fixed ε returns N_{eff} unchanged at the level of the code's output [5]. The deformation is therefore a pure rate effect, not an expansion-rate lever.

III. MEASURED HELIUM RESPONSE

The bound is built from a production run of the PRyMordial nuclear network [5], with the ramp of Eq. (2) applied pointwise to the weak rates: at each temperature the standard and fully shifted rates are blended by the weight $w(T) = \max(0, 1 - T/T_c)$. The run that defines the response was performed at

$$T_c = 179 \text{ keV}, \quad \varepsilon = 1.2543\%, \quad (4)$$

on the code's default baryon density. The helium mass fraction moves from the $\varepsilon = 0$ baseline to the windowed value as

$$Y_p^{(0)} = 0.246891 \rightarrow Y_p^{(\varepsilon)} = 0.248995, \quad (5)$$

a fractional change of $+0.852\%$.

A four-point scan in ε at the same T_c establishes that the helium response is linear to a few percent over the range of interest, with elasticity

$$\frac{dY_p}{d\varepsilon} = 0.00163 \quad \text{per percent of } \varepsilon. \quad (6)$$

Multiplying Eq. (6) by $\varepsilon = 1.2543\%$ recovers the window of Eq. (5) to 2.8% of the window itself. The linearity is what licenses a derivative bound; without it the translation from one windowed run into a limit on ε would not be legitimate.

Helium is almost insensitive to the baryon density, $Y_p \propto \omega_b^{0.04}$. Shifting ω_b at the percent level therefore moves the baseline by far less than the observational error used below, so the bound is baseline-robust against the usual CMB–BBN ω_b re-inference.

IV. THE AVER BOUND

Aver *et al.* report [6]

$$Y_p^{\text{Aver}} = 0.2453 \pm 0.0034. \quad (7)$$

Writing $Y_p(\varepsilon) = Y_p^{(0)} + 0.00163 \varepsilon\%$ with $\varepsilon\%$ the amplitude in percent, and requiring $Y_p(\varepsilon)$ to lie within $n\sigma$ of Eq. (7), one finds

$$\varepsilon < 1.11\% \quad (1\sigma), \quad (8)$$

$$\varepsilon < 3.20\% \quad (2\sigma). \quad (9)$$

The 2σ statement is the result of this paper:

$$\boxed{\varepsilon < 3.2\% \quad (2\sigma)} \quad (10)$$

against Aver, at the T_c of Eq. (4), with no fitted parameters.

For orientation: the $\varepsilon = 0$ baseline already sits $+0.47\sigma$ above Aver, and the windowed point of Eq. (4) sits at $+1.09\sigma$. The transition is constrained but not excluded at the amplitude of Eq. (4). The 1σ column of Eq. (8) is reported for completeness; the quotable bound is Eq. (10).

A dense map of the upper edge $\varepsilon_{\text{max}}(T_c)$ over the full interval of Eq. (3) is the natural extension of Eq. (10). It requires a grid of network runs in T_c at fixed observational Y_p , which is not reported here. The number in Eq. (10) is the verified bound at the measured T_c ; it is not a claim that the same numerical ceiling holds at every point of Eq. (3).

V. WHY EMPRESS CANNOT BE USED

The EMPRESS collaboration reports a lower helium abundance [7],

$$Y_p^{\text{EMPRESS}} = 0.2370 \pm 0.0034. \quad (11)$$

Against that determination the same baseline and windowed values read

$$\varepsilon = 0 : +2.91\sigma, \quad \varepsilon = 1.2543\% : +3.53\sigma. \quad (12)$$

Standard BBN with no transition is already 2.9σ high. The window contributes only 0.62σ of the 3.53σ . A measurement that is in tension with the null hypothesis cannot constrain a deformation of that null: removing the transition entirely leaves 2.9σ standing. Quoting an EMPRESS-based upper limit on ε would attribute to the transition a discrepancy that exists without it.

The observational helium disagreement itself — Aver and EMPRESS differ by about 1.7σ on the same quantity — is a property of the data, not of the ramp. Until it is resolved, a helium-only bound on ε must be reported against Aver and must state why EMPRESS is not used. That is the entire content of this section.

VI. WHAT IS NOT USED

Deuterium. The deuterium abundance responds to ε , but as a bottleneck residue rather than an equilibrium abundance. A uniform- ε derivative under-predicts the windowed effect, and a four-point scan does not yield a linear elasticity that can be inverted into a clean upper limit of the kind in Eq. (10). No derivative-based deuterium bound is claimed. Absolute D/H comparisons that depend on a re-inferred baryon density from a cosmological fit are outside the scope of this note: they are not needed for Eq. (10), and they are not used.

Extra radiation. An additional relativistic density ΔN_{eff} moves Y_p and D/H in the same direction. It is an expansion-rate lever, not a measurement of ε , and it is not floated here. The bound of Eq. (10) is the pure rate response at fixed standard N_{eff} .

Lithium. The same windowed run moves ${}^7\text{Li}/\text{H}$ by +0.26%, a percent-level shift that neither causes nor cures the cosmological lithium problem. Lithium is recorded as unaffected at the level relevant to Eq. (10) and is not used in the bound.

Model-fixed T_c or ε . No central value of T_c or ε is adopted as an input. The run of Eq. (4) fixes the elasticity; the bound is the upper edge that Aver allows.

VII. CONCLUSION

A universal electron-mass shift that turns on inside the BBN window, with free amplitude ε and free critical temperature T_c in the nucleosynthesis-relevant range, is bounded by primordial helium. Against Aver *et al.*, and at the T_c where the helium elasticity was measured,

$$\varepsilon < 3.2\% \quad (2\sigma),$$

with zero fitted parameters. The EMPRESS determination cannot be used for this purpose, because standard BBN is already 2.9σ high against it before any transition is applied. Deuterium is excluded from the bound because its ε -response is nonlinear. The result is a chain-free, single-observable upper limit on the amplitude of a leptonic electron-mass transition inside the nucleosynthesis window.

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